Predicting Student Performance with Gradient Boosted Trees

XGBoost Baseline vs. CatBoost Alternative

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Problem
& Motivation

- Goal: predict final course grade category (A–F) from demographic, support, and prior performance features.
- Motivation: identify at–risk students early to support interventions.
- Challenges:
  - Strong class imbalance (majority in top grade category).
  - Many categorical features and non–linear relationships.

Dataset

- Source: Students Performance Dataset from Kaggle[1]
- Observations: several hundred students (one row per student).
- Features:
  - Demographic: gender, ethnicity, parental education.
  - Support: tutoring, parental support, extracurriculars, sports, music, volunteering.
  - Performance: prior scores, absences, etc.
- Target:
  - Numeric grade mapped to 5 ordered categories (A, B, C, D, F).
  - Encoded as integer labels 0–4 via a fixed grade map.

[1] Rabie El Kharoua, "Students Performance Dataset," Kaggle,
https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset.

Class Imbalance

- Grade distribution is highly skewed:
  - Majority of students in the highest grade category.
  - Relatively few students in A and B categories.
- Consequences:
  - Accuracy alone can be misleading.
  - Models may ignore minority grades yet appear to perform well.
- Design choice: focus on macro–averaged F_1 instead of only accuracy.

Problem Setup & Metrics

- Task: 5–class classification of `GradeClass` (0=A, 1=B, 2=C, 3=D, 4=F).
- Splits:
  - 20% stratified held–out test set.
  - Remaining 80% split into train and validation (stratified, same random seed).
- Primary evaluation metric:
  - Macro–averaged F_1 on held–out test set:

$$F_1^{\text{macro}} = \frac{1}{K} \sum_{k=1}^{5} F_1^{(k)}.$$

- Secondary metrics: accuracy, per–class precision/recall/F_1, confusion matrices.

Features & Preprocessing

- Dropped features to avoid leakage: `StudentID`, `GPA`.
- Categorical features:
  - Gender, Ethnicity, ParentalEducation, Tutoring, ParentalSupport, Extracurricular, Sports, Music, Volunteering.
- Preprocessing (shared for both models):
  - Categorical columns cast to categorical type.
  - Remaining columns coerced to numeric.
  - All models run on the same feature matrix and label vector.

Baseline Model: XGBoost

- Model: `XGBClassifier` (gradient boosted decision trees)[2]
- Key hyperparameters:
  - Objective: `multi:softprob`; tree method: `hist`.
  - Max depth = 6, learning rate = 0.1, 500 trees.
  - Subsample = 0.9, colsample extunderscore bytree = 0.9, $\lambda = 1.0$.
- Training:
  - Early stopping on validation set with multi–class log–loss.
  - 5–fold stratified cross–validation on training split using macro F_1.

[2] Chen, T.,
Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD. arXiv:1603.02754.

Alternative Model: CatBoost

- Model: `CatBoostClassifier`[3]
- Motivation:
  - Native handling of categorical features.
  - Built–in support for class imbalance via `auto_class_weights`.
- Key hyperparameters:
  - Loss: `MultiClass`; eval metric: macro F_1.
  - Depth $= 6$, learning rate $= 0.1$, 500 iterations.
  - L2 regularization $= 3.0$, `auto_class_weights = "Balanced"`.
- Training:
  - Same train/validation split as XGBoost.
  - 5–fold stratified cross–validation on training split.

[3] Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., Gulin, A. (2018). CatBoost: unbiased boosting with categorical features. NeurIPS. arXiv:1706.09516.

Cross–Validation Results

Model	CV macro F_1 (mean)	CV macro F_1 (std)
XGBoost	0.566	0.057
CatBoost	0.606	0.035

- CatBoost achieves higher average macro F_1 with slightly lower variance.
- Suggests better generalization across folds, especially for minority classes.

Held–Out Test Results

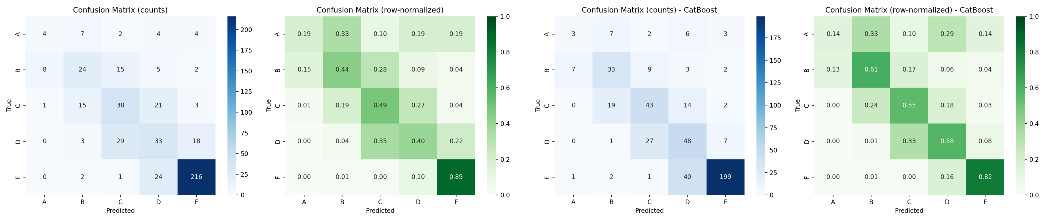
Model	Accuracy	Macro F_1
XGBoost	0.658	0.487
CatBoost	0.681	0.532

- CatBoost improves both overall accuracy and macro F_1 on the test set.
- Improvement in macro F_1 indicates better handling of minority grades.

Confusion Matrices

XGBoost

CatBoost



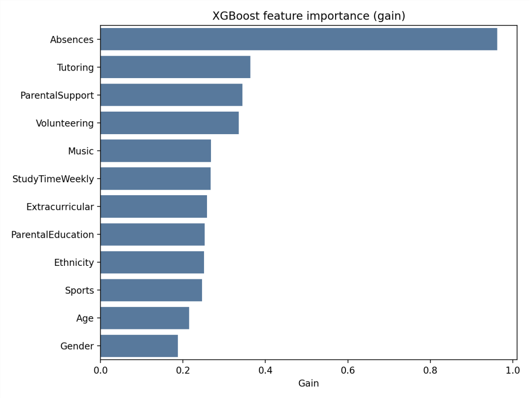
- XGBoost tends to over–predict the majority grade.
- CatBoost distributes predictions more across B, C, and D, boosting recall.

Per–Class Performance (F_1)

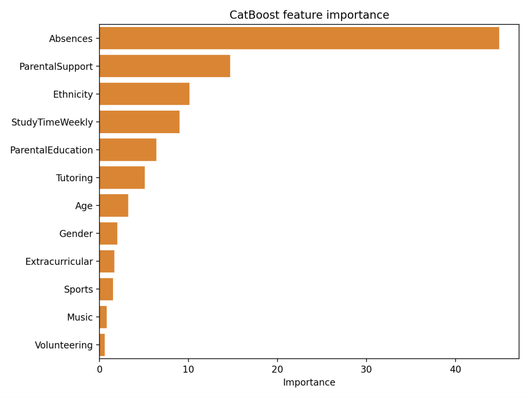
- Selected F_1 scores (XGBoost $\rightarrow$ CatBoost):
  - Grade B: $0.46 \rightarrow 0.57$
  - Grade C: $0.47 \rightarrow 0.54$
  - Grade D: $0.39 \rightarrow 0.49$
- Largest gains occur for mid–range grades, which are most challenging to classify.
- Failing grade remains easy to detect for both models due to distinct patterns.

Feature Importance

XGBoost (gain)



CatBoost



- Both models highlight prior performance, tutoring, and parental support as key signals. CatBoost places slightly more emphasis on categorical support variables.

Example Predictions

- Three illustrative test cases (high–confidence predictions):

Index	True Grade	Predicted Grade	Confidence
475	B	A	0.999
118	A	B	0.998
464	F	F	1.000

- Boundary errors (A vs. B) are common and reflect overlapping feature patterns.
- Failing students are predicted very confidently and accurately.

Limitations

- Data:
  - Modest dataset size; limited diversity of features.
  - No temporal/behavioral sequence information.
- Modeling:
  - Only modest hyperparameter tuning; no large search.
  - Single train/test split (though cross–validation used on train).
- Interpretability:
  - Tree ensembles are complex; feature importance is only a coarse explanation.
  - No calibration of predicted probabilities.

Future Work

- Expanded hyperparameter tuning for both XGBoost and CatBoost.
- Explore additional models (LightGBM, calibrated linear models, ensembles).
- Try alternative imbalance strategies: custom class weights, oversampling, focal loss.
- Incorporate richer features (attendance trends, assignment histories, engagement data).
- Calibrate probabilities and design decision thresholds focused on at–risk students.

Summary

- Built an end–to–end pipeline: EDA, preprocessing, XGBoost baseline, CatBoost alternative.
- CatBoost improves cross–validated macro F_1 ($0.57 \to 0.61$) and test macro F_1 ($0.49 \to 0.53$), with slightly higher accuracy.
- Gains are largest for mid–range grades (B, C, D), which matter for identifying borderline students.
- Results highlight the value of models that respect categorical structure and class imbalance in educational prediction tasks.

References

- Rabie El Kharoua. “Students Performance Dataset.” Kaggle. `https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset`.
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD. arXiv:1603.02754.
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: unbiased boosting with categorical features. NeurIPS. arXiv:1706.09516.